

Measuring Song Success

Grant Z, Haoting Y, Max E, Max VZ

Meet the Team



Grant Zou

Fav Album:
-Glass Houses by Billy Joel



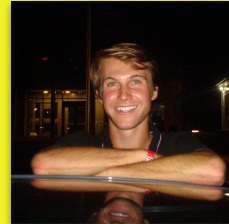
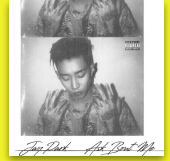
Max VZ

Fav Album:
- Led Zeppelin IV
by Led Zeppelin



HY

Fav Album:
-Ask About Me by Jay Park



Max E

Fav Album:
- When We Were Friends
by The Backseat Lovers



\$46 Billion*

Global Music Streaming Market Size Estimate 2024

*Grand View Research

\$108 Billion*

Global Streaming Market Projected Value in 2030

*Grand View Research

Background and Motivation

A DJ wearing a VR headset is performing at a DJ booth in a nightclub. The scene is dimly lit with vibrant blue and purple neon lights. The DJ is wearing a light-colored button-down shirt and is focused on the VR headset while their hands are on the DJ mixer. In the background, other patrons of the club are visible, some also wearing VR headsets, creating a futuristic atmosphere.

What does the music industry look like?
What are “the charts”? Why do they matter?
How have people analyzed music data in the past?
Why is this an interesting and novel area of research?

01

Motivation



Music listeners are moving towards streaming

Streaming is now the most popular way for people to listen to music, overtaking radio and physical records



Album cover art is prominent in streaming services

Little research has been done into this area. We want to determine if music producers should increase the resources that go into album cover art design

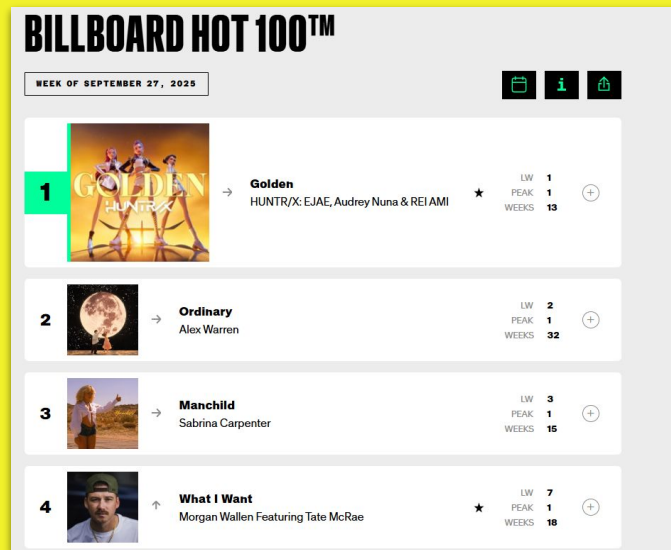


Audio characteristics

We will also investigate audio characteristics to find additional relationships with cover art

- A chart of the Top 100 songs in the United States published by Billboard magazine every week
 - Ranked based on streams, radio plays, physical sales
- Breaking into the Top 100 makes a name for artists
- Despite major changes to the music industry with the advent of streaming, creating songs which enter the Billboard Top 100 remains a goal of all artists

What is the Billboard Hot 100?



Statistical Background

Stanford (Billboard)

76.5%*

- Neural Network
- Logistic Regression (75.9%)
- GDA (73.7%)

Key Findings

- Valence (happiness) matters
- High energy reduces survival
- Featured guests improve performance

Claremont (Spotify)

45.5% γ

- OLS Adjusted R-squared
- Peak position and survival time across countries

Common Features

10+

Danceability, Valence, Energy, Tempo, Acousticness, etc.

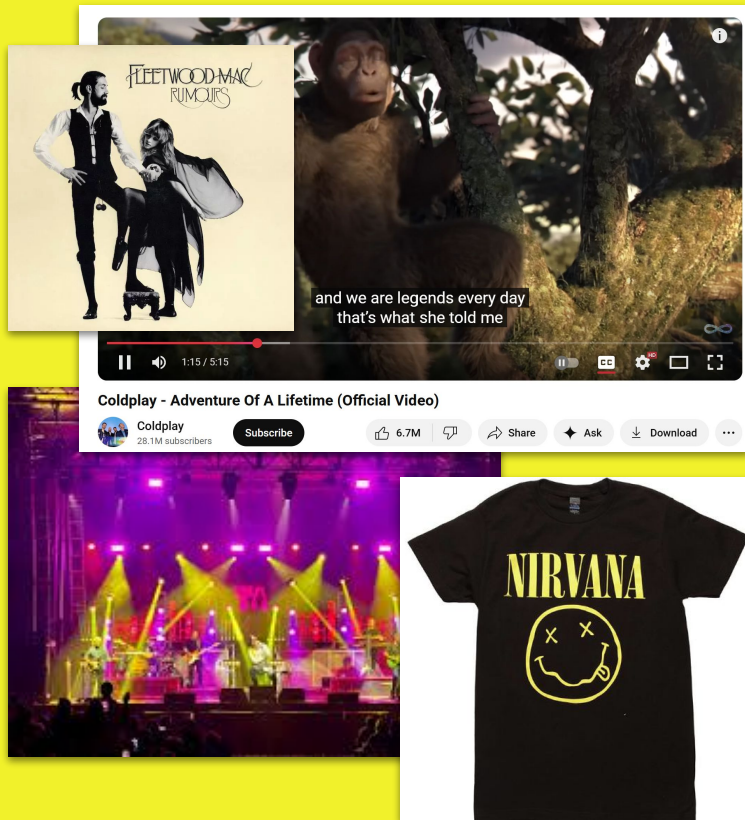
*Stanford γ Claremont

10+ (Audio)

Common Features Across Literature

Danceability, Valence, Energy, Tempo, Acousticness, etc.

New Territory



Multi-Sensory Experience

Music is a multi-sensory experience. Through live shows with lights and backgrounds, album covers, music videos, merchandise, and more, artists build not only a piece of music, but a brand.



Visual + Audio

- Color palettes and visual complexity
- Typography and graphic elements
- Beyond genre classification to prediction

Visual Component

A person wearing a VR headset is seen from behind, looking out over a large, vibrant nightclub scene at night. The person is wearing a camouflage jacket and has their hair tied back. The nightclub is filled with people, colorful lights, and large screens displaying abstract visuals. The overall atmosphere is energetic and immersive.

What is it?
Why does it matter?
How can we analyze it?

02

- **Began as vinyl record packaging**
- **Visual branding**
- **Artistic Expression**
- **Merchandising and culture**



What is Cover Art?

How is Cover Art Made?



Pink Floyd's 'The Dark Side of the Moon' was designed by the design group Hipgnosis. Now it is one of the most recognizable cover arts of all time



The Musical Artist

Musical artists usually play a large role in the creation of album cover art. If the artist is visually creative they may design their own.



Contracted Independent Artists

Otherwise they will work with independent artists.

- **Changing Listening Landscape**
 - Interacting with songs in new ways
 - 84% correlation between TikTok & Billboard Top 200
- **YouTube**
 - Music videos & fan engagement
- **Intertwined song experience**
 - Rise of mobile streaming

Increased Impact of Visuals



How does this increase generate revenue?



- Sony reported over 1 billion dollar increase in Music sales from 2023 to 2024
- Commercial Ad Revenue from music videos
- Premium streaming services
- Over 40% of global music engagement directly involves a visual aspect

Metrics into money?

How will we analyze cover art?

1. Memorability is stable and predictable

Across different viewers, the probability an image will be remembered is highly consistent; predicting memorability from image features is feasible.¹

2. Memorability can be modeled from image content

Global descriptors (e.g., GIST/HOG/SIFT) and semantic cues (objects & scenes) reach solid rank correlations, showing content-level signals (not just pixel stats) matter.¹

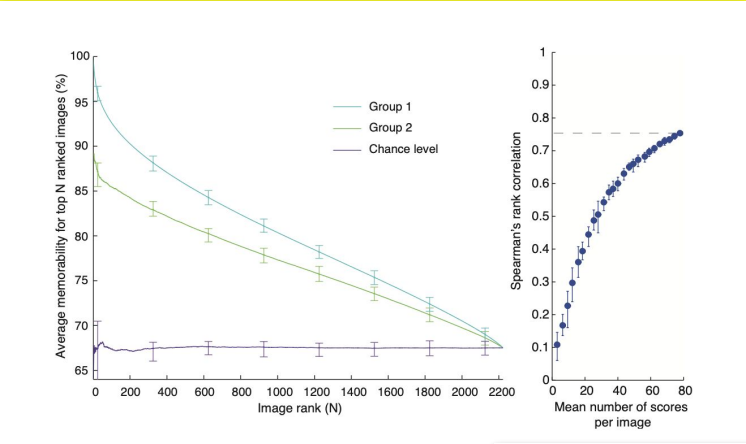
3. Covers encode learnable design rules

A CNN trained only on cover images can predict 30 book genres (Top-1 ≈24.7%, Top-3 ≈40.3%), revealing recurring visual/textual patterns (e.g., big sans-serif titles → Thriller; close-up faces → Biographies).²

Visual elements have a measurable “ground truth”

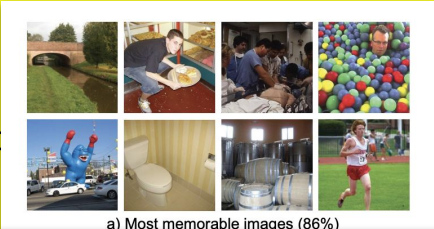
1. Memorability is stable and predictable

Across different viewers, the probability an image will be remembered is **remembered** is **highly consistent**; predicting memorability from image features is feasible.¹



from image content

(T) and semantic cues (objects & scene labels (not just pixel stats) matter.¹



a) Most memorable images (86%)

rules

n predict 3
g sans-se

	Pixels	GIST	SIFT	SSIM	HOG 2x2	All Global Features
Top 20	74%	82%	83%	83%	83%	83%
Top 100	72%	78%	79%	79%	80%	80%
Bottom 100	61%	58%	57%	58%	57%	56%
Bottom 20	59%	57%	56%	56%	55%	54%

	Object Counts	Object Areas	Multiscale Object Areas	Object Label Presences	Labeled Object Counts	Labeled Object Areas	Labeled Multiscale Object Areas	Scene Category	Objects and Scenes	Other Humans
Top 20	68%	67%	73%	84%	82%	84%	84%	81%	85%	86%
Top 100	68%	68%	73%	79%	79%	82%	82%	78%	82%	84%
Bottom 100	67%	64%	64%	57%	57%	56%	56%	57%	55%	47%
Bottom 20	67%	63%	65%	55%	54%	53%	52%	55%	53%	40%
ρ	0.05	0.05	0.20	0.43	0.44	0.47	0.48	0.37	0.50	0.75

Table 1. Comparison of predicted versus measured memorabilities. Images are sorted into sets according to predictions made on the basis of a variety of object and scene features (denoted by column headings). Average empirically measured memorabilities are reported for each set. e.g. The “Top 20” row reports average empirical memorability over the images with the top 20 highest predicted memorabilities. ρ is the Spearman rank correlation between predictions and measurements.

Visual elements
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1. Pixel-level feature complexity → inverted-U with likes

Color variation, luminance, and edge density show clear sweet spots: too simple under-informs; too busy overwhelms. Evidence from ~150k Instagram brand posts with negative-binomial models.³

2. Design complexity matters—mind symmetry and arrangement

Number of unique objects & irregularity show U-shapes; asymmetry is monotonically negative for liking across the observed range.³

3. Fashionability can be predicted—and improved—with recommendations

A CRF that jointly reasons about outfit (visual content), user, scene, and metadata predicts fashionability (votes/likes) and outputs actionable suggestions (e.g., change outfit/setting) on a 144k-post social dataset.⁴

Visual complexity & fashionability imply engagement

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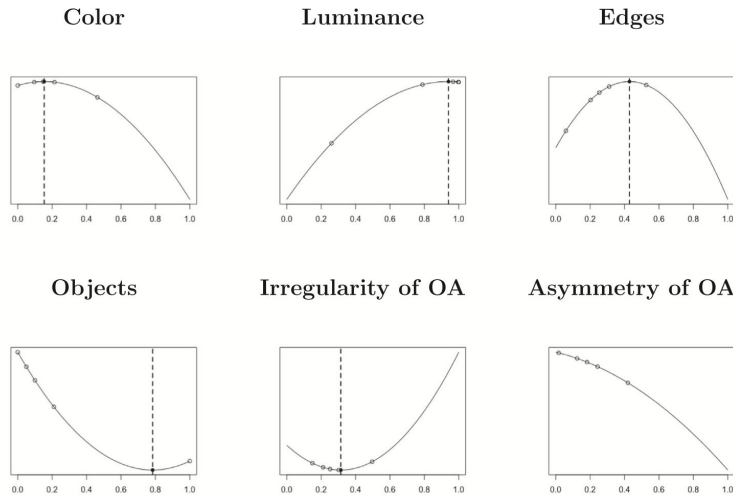


Fig. 4. Visualization of the effects for each individual measure. The variables in the regression analysis are normalized to be between 0 and 1. The y-axis represents the estimated number of likes, all else being equal. The circles on the plot represent the percentiles (1%, 25%, 50%, 75%, 99%) of the distribution of the independent variable. The dashed line and the solid point indicate the global maximum/minimum.



Fig. 1. Sample images of low, medium and high complexity for each individual measure of feature complexity.

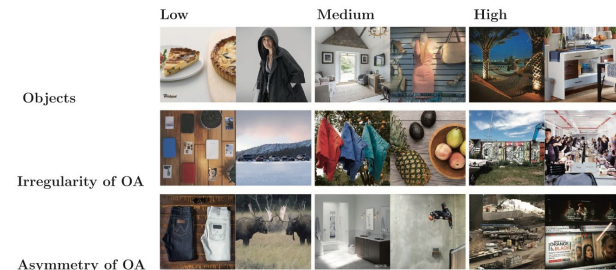
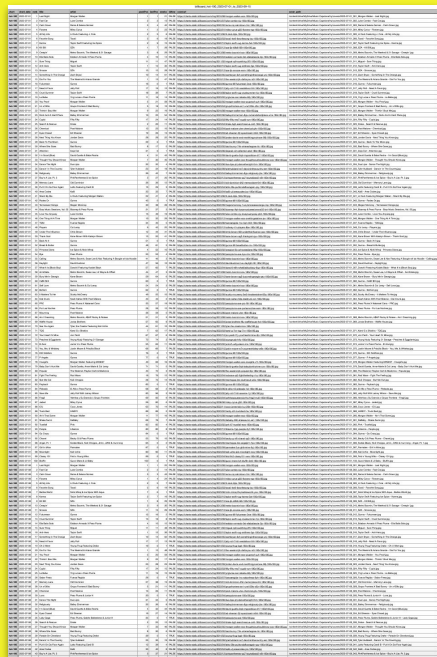
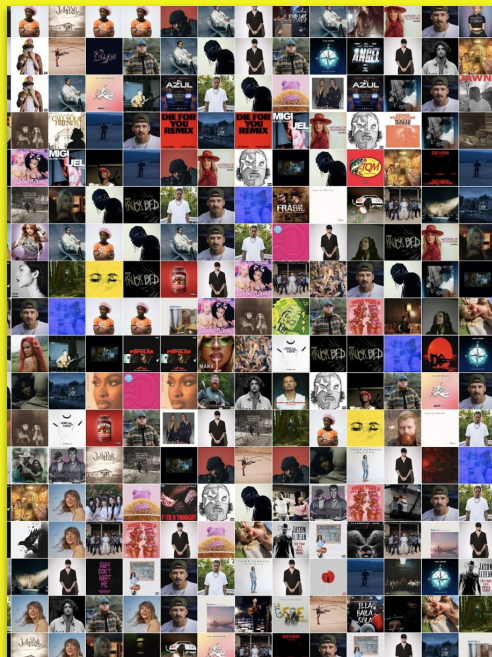


Fig. 2. Sample images of low, medium and high complexity for each individual measure of design complexity. Number Objects, Irregularity of Object Arrangement, Asymmetry of Object Arrangement.

From [billboard.py](#), we get billboard covers.
From billboard covers, we extract visual features.
+ More features to be explored



Rank	Artist	Album	Label	Release Date	Genre	Chart Position	Weeks on Chart	Peak Position	Current Position
1	Drake	For All the Dogs	OVO Sound	2023-10-27	Hip-Hop/Rap	1	1	1	1
2	Travis Kelce	Travis Kelce	Various	2023-10-27	Pop	2	2	2	2
3	Drake	For All the Dogs	OVO Sound	2023-10-27	Hip-Hop/Rap	3	3	3	3
4	Drake	For All the Dogs	OVO Sound	2023-10-27	Hip-Hop/Rap	4	4	4	4
5	Drake	For All the Dogs	OVO Sound	2023-10-27	Hip-Hop/Rap	5	5	5	5



Data & Analysis Roadmap

The stage has been set...



Studies have investigated song audio features in attempts to predict success, but there has been limited exploration of visual aspects.

We will look into measuring and predicting song success with a focus on cover art alongside other features.

Moving Forward ~

THANK YOU!

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